DRAFT: The Scalar-Angular-Torsion Framework: VIII. Falsification (The "Honesty Test")

Nathan McKnight
Independent Researcher
(Dated: March 11, 2026)

We conclude the initial formalization of the Scalar-Angular-Torsion (SAT) framework by defining the experimental parameters for its falsification. We characterize the Achromatic Phase Snap—a discrete, frequency-independent jump in timing residuals—as the primary "Smoking Gun" for the discretized 24-cell HSUCV lattice. This phenomenon is predicted to manifest at the Critical Velocity threshold ($v_{crit} \approx 0.2387c$). Finally, we establish the "Hard STOP" conditions, identifying specific empirical observations that would necessitate the immediate rejection of the SAT ontology.

LAY OVERVIEW

The SAT framework is designed to be "fail-rigid." Unlike many theories that can be adjusted with "fudge factors" to fit new data, this model is built so that it either fits the universe exactly or it breaks completely. This final section provides the experimental "Smoking Gun": the **Achromatic Phase Snap**. Because the universe is built on a 4D grid (the lattice), any object moving fast enough will eventually have to "jump" or "click" from one grid point to the next. We predict this jump happens at exactly 71,500 km/s and results in a sudden, discrete shift of 14.1 degrees in the object's timing.

This section also sets the **Hard STOP Conditions**. These are the specific scientific observations that would immediately prove the SAT framework is false. For example, if we ever find a stable particle with a "knot complexity" higher than three, or if the predicted 14.1-degree shift fails to happen at the exact speed predicted, the theory is dead. This "Honesty Test" ensures that the framework remains a tool of rigid science rather than a flexible mathematical exercise.

I. THE ACHROMATIC PHASE SNAP

The definitive experimental signature of the Zottenwelt is the **Achromatic Phase Snap**. While 4D filaments move continuously, their geodesic paths are constrained by the discrete nodes of the 24-cell HSUCV lattice. This discretization implies that worldline rotation is not smooth at relativistic thresholds; instead, the worldline must suddenly "click" or "snap" to the next lattice node to maintain topological stability.

This transition is predicted to occur at the **Critical Velocity (v_{crit})**, which is the mechanical product of the Projection Constant B and the expansion rate c:

$$v_{crit} = B \cdot c \approx 0.2387c \approx 71,500 \text{ km/s} \tag{1}$$

At this threshold, the filament tangent reaches the **Obscuration Constant** ($\theta_{obs} \approx 0.246$ rad). The resulting phase snap is characterized by a discrete, frequency-independent jump in the timing residuals of the system, precisely equal to the Obscuration Constant:

$$\Delta\phi_{residual} = 0.246 \text{ rad } (14.1^\circ) \tag{2}$$

This "staccato" motion would be absent in any continuous 3D spacetime model, serving as an unambiguous marker of 4D lattice geometry.

II. HARD STOP CONDITIONS

The structural integrity of the SAT framework is mortgaged on the principle of **Failure Rigidity**. The following observations constitute "Hard STOP" conditions that would fundamentally falsify the theory:

1. **High-Complexity Stability:** The discovery of any stable ground-state particle possessing a topological charge $Q \geq 4$. This would violate the **UV Finiteness Lock** that caps intersections at three filaments per node.

2. **Scaling Elasticity:** Evidence that the B^4 scaling used for lepton mass is a manual fit rather than a mandatory requirement derived from 24-cell lattice packing.

3. **Phase Snap Failure:** The failure of the 0.246 rad achromatic phase shift to manifest at the precise $0.2387c$ velocity threshold.

III. METROLOGICAL VERDICT

By anchoring all physical laws to a single geometric scale ($l_f \approx 0.7937$ fm) and the rigid architecture of the 24-cell lattice, the SAT framework eliminates the "geometric expense" of tuned parameters. The framework is self-diagnostic; any error in the fundamental anchors ripples catastrophically through every derived sector. This rigidity ensures that the SAT ontology is either a total structural match to the universe or an immediately falsifiable simulation.

CONCLUSION

The Scalar-Angular-Torsion framework represents a transition toward a physics of rigidity and honesty. By reinterpreting mass as projective resistance, gravity as collective attenuation, and particles as topological traces, we provide a unified coordinate engine capable of resolving the Standard Model's arbitrary constants into geometric inevitabilities.